

# Customer Trust & Engagement Segmentation using Ratings Data

## Project Overview

This phase of the project analyzed customer ratings and review volumes to understand how product categories perform in terms of both customer trust (satisfaction) and market engagement (demand).

A Trust Score was created by combining customer ratings with the number of ratings to distinguish between categories that are:

- High demand and high satisfaction
- High demand but low satisfaction
- Low demand but high satisfaction
- Low demand and low satisfaction

Using this framework, categories were segmented into four strategic groups:

- High Engagement – High Ratings: Stable, high-performing categories with strong product-market fit and predictable revenue potential.
- High Engagement – Low Ratings: Categories with strong demand but weak customer experience, indicating execution or quality gaps that require operational fixes.
- Low Engagement – High Ratings: Niche or underexposed categories with good customer satisfaction but low visibility, representing demand-generation opportunities.
- Low Engagement – Low Ratings: Categories with weak demand and poor satisfaction, signaling areas where further investment may harm brand trust.

This analysis reframed ratings as a business decision signal, helping identify where to scale, where to fix product issues, where to invest in visibility, and where to reduce promotion.

## Data Set Summary

### Dataset Context

The dataset, Amazon Sales Data 2023, contains detailed information on 551,585 products across 19 main categories and 112 sub-categories.

It includes 9 columns covering product details, pricing, customer ratings, and access information.

### Key Features

- Product details: Name, Main Category, Sub-category
- Pricing: Discount Price, Actual Price
- Ratings: Average Rating, Number of Ratings
- Media & access: Image URL, Product Link

This dataset supports large-scale marketplace analysis across multiple product categories.

### 3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Loaded the CSV file into a DataFrame and previewed the first 15 records to understand the data layout and contents.

|   | name                                               | main_category | sub_category     | image                                                                                                             | link                                                                                                                  | ratings | no_of_ratings | discount_price | actual_price |
|---|----------------------------------------------------|---------------|------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------|---------------|----------------|--------------|
| 0 | Lloyd 1.5 Ton 3 Star Inverter Split AC (5 In 1...  | appliances    | Air Conditioners | <a href="https://m.media-amazon.com/images/I/31UISB90sY...">https://m.media-amazon.com/images/I/31UISB90sY...</a> | <a href="https://www.amazon.in/Lloyd-Inverter-Convertible...">https://www.amazon.in/Lloyd-Inverter-Convertible...</a> | 4.2     | 2255.0        | 32999.0        | 58990.0      |
| 1 | LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (C...  | appliances    | Air Conditioners | <a href="https://m.media-amazon.com/images/I/51JFb7FctD...">https://m.media-amazon.com/images/I/51JFb7FctD...</a> | <a href="https://www.amazon.in/LG-Convertible-Anti-Viru...">https://www.amazon.in/LG-Convertible-Anti-Viru...</a>     | 4.2     | 2948.0        | 46490.0        | 75990.0      |
| 2 | LG 1 Ton 4 Star AI Dual Inverter Split AC (Copp... | appliances    | Air Conditioners | <a href="https://m.media-amazon.com/images/I/51JFb7FctD...">https://m.media-amazon.com/images/I/51JFb7FctD...</a> | <a href="https://www.amazon.in/LG-Inverter-Convertible-...">https://www.amazon.in/LG-Inverter-Convertible-...</a>     | 4.2     | 1206.0        | 34490.0        | 61990.0      |
| 3 | LG 1.5 Ton 3 Star AI DUAL Inverter Split AC (C...  | appliances    | Air Conditioners | <a href="https://m.media-amazon.com/images/I/51JFb7FctD...">https://m.media-amazon.com/images/I/51JFb7FctD...</a> | <a href="https://www.amazon.in/LG-Convertible-Anti-Viru...">https://www.amazon.in/LG-Convertible-Anti-Viru...</a>     | 4.0     | 69.0          | 37990.0        | 68990.0      |
| 4 | Carrier 1.5 Ton 3 Star Inverter Split AC (Copp...  | appliances    | Air Conditioners | <a href="https://m.media-amazon.com/images/I/41lrtqXPW...">https://m.media-amazon.com/images/I/41lrtqXPW...</a>   | <a href="https://www.amazon.in/Carrier-Inverter-Split-C...">https://www.amazon.in/Carrier-Inverter-Split-C...</a>     | 4.1     | 630.0         | 34490.0        | 67790.0      |

- **Standardized numeric fields** by removing currency symbols, commas, and text to ensure consistent numerical formatting.
- **Converted relevant columns to numeric data types** to make them suitable for calculations and analysis.
- **Removed duplicate products** using a defined business key to avoid double counting in analysis.
- **Renamed columns for consistency and clarity**, improving readability and downstream usability.

### 4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

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```

select main_category ,
       sub_category ,
       ROUND(AVG(Ratings * LOG(no_of_ratings)), 2 ) AS Trust_Score_subcat
from amazon_database
GROUP BY main_category , sub_category
order by trust_score_subcat DESC
limit 10;

```

|    | main_category<br>text    | sub_category<br>text         | trust_score_subcat<br>numeric |
|----|--------------------------|------------------------------|-------------------------------|
| 1  | beauty & health          | Value Bazaar                 | 11.79                         |
| 2  | toys & baby products     | International Toy Store      | 10.49                         |
| 3  | toys & baby products     | Toys Gifting Store           | 9.74                          |
| 4  | beauty & health          | Health & Personal Care       | 9.54                          |
| 5  | grocery & gourmet foo... | All Grocery & Gourmet Foo... | 9.16                          |
| 6  | tv, audio & cameras      | All Electronics              | 8.71                          |
| 7  | home & kitchen           | All Home & Kitchen           | 8.64                          |
| 8  | stores                   | Amazon Fashion               | 8.14                          |
| 9  | toys & baby products     | Baby Products                | 7.93                          |
| 10 | beauty & health          | Beauty & Grooming            | 7.81                          |

```

WITH low_engagement AS (
    SELECT
        |   main_category,
        |   sub_category,
        |   ROUND(
        |       PERCENTILE_CONT(0.5)
        |       WITHIN GROUP (ORDER BY no_of_ratings)::numeric,
        |       2
        |   ) AS no_of_ratings_median
    FROM amazon_database
    GROUP BY main_category, sub_category
    -- ORDER BY no_of_ratings_median DESC
    LIMIT 50
),
low_rating AS (
    SELECT
        main_category,
        sub_category,
        ROUND(
            PERCENTILE_CONT(0.5)
            WITHIN GROUP (ORDER BY RATINGS)::numeric,
            2
        ) AS median_rating_subcat
    FROM amazon_database
    GROUP BY main_category, sub_category
    -- ORDER BY median_rating_subcat ASC
    LIMIT 50
)
SELECT
    low_engagement.sub_category,
    low_engagement.main_category,
    low_engagement.no_of_ratings_median,
    low_rating.median_rating_subcat
FROM low_engagement
INNER JOIN low_rating
    ON low_engagement.sub_category = low_rating.sub_category
order by median_rating_subcat asc , no_of_ratings_median asc
limit 10
;

```

|    | main_category<br>text    | sub_category<br>text         | no_of_ratings_median<br>numeric | median_rating_subcat<br>numeric |
|----|--------------------------|------------------------------|---------------------------------|---------------------------------|
| 1  | beauty & health          | Value Bazaar                 | 450.00                          | 4.30                            |
| 2  | toys & baby products     | International Toy Store      | 400.00                          | 4.60                            |
| 3  | toys & baby products     | Toys Gifting Store           | 315.00                          | 4.15                            |
| 4  | beauty & health          | Health & Personal Care       | 281.00                          | 4.20                            |
| 5  | grocery & gourmet foo... | All Grocery & Gourmet Foo... | 221.00                          | 4.20                            |
| 6  | tv, audio & cameras      | All Electronics              | 214.00                          | 4.10                            |
| 7  | home & kitchen           | All Home & Kitchen           | 208.00                          | 4.10                            |
| 8  | stores                   | Amazon Fashion               | 157.00                          | 4.10                            |
| 9  | beauty & health          | Beauty & Grooming            | 119.00                          | 4.10                            |
| 10 | home & kitchen           | Kitchen & Dining             | 119.00                          | 4.10                            |

```

WITH high_engagement AS (
    SELECT
        main_category,
        sub_category,
        ROUND(
            PERCENTILE_CONT(0.5)
            WITHIN GROUP (ORDER BY no_of_ratings)::numeric,
            2
        ) AS no_of_ratings_median
    FROM amazon_database
    GROUP BY main_category, sub_category
    ORDER BY no_of_ratings_median DESC
    LIMIT 50
),
low_rating AS (
    SELECT
        main_category,
        sub_category,
        ROUND(
            PERCENTILE_CONT(0.5)
            WITHIN GROUP (ORDER BY ratings)::numeric,
            2
        ) AS median_rating_subcat
    FROM amazon_database
    GROUP BY main_category, sub_category
    ORDER BY median_rating_subcat ASC
    LIMIT 50
)
SELECT
    high_engagement.sub_category,
    high_engagement.main_category,
    high_engagement.no_of_ratings_median,
    low_rating.median_rating_subcat
FROM high_engagement
INNER JOIN low_rating
    ON high_engagement.sub_category = low_rating.sub_category
ORDER BY
    high_engagement.no_of_ratings_median DESC,
    low_rating.median_rating_subcat ASC
LIMIT 10;

```

|    | sub_category<br>text             | main_category<br>text | no_of_ratings_median<br>numeric | median_rating_subcat<br>numeric |
|----|----------------------------------|-----------------------|---------------------------------|---------------------------------|
| 1  | Industrial & Scientific Suppl... | industrial suppli...  | 70.00                           | 4.00                            |
| 2  | All Sports, Fitness & Outdoo...  | sports & fitness      | 64.00                           | 3.90                            |
| 3  | All Car & Motorbike Products     | car & motorbike       | 55.00                           | 3.90                            |
| 4  | Women's Fashion                  | stores                | 54.50                           | 3.90                            |
| 5  | Kitchen & Home Appliances        | appliances            | 49.00                           | 3.90                            |
| 6  | Men's Fashion                    | stores                | 46.00                           | 3.90                            |
| 7  | Clothing                         | women's clothing      | 42.00                           | 3.80                            |
| 8  | Fashion Sales & Deals            | stores                | 32.00                           | 3.70                            |
| 9  | Garden & Outdoors                | home & kitchen        | 31.00                           | 3.90                            |
| 10 | Running                          | sports & fitness      | 26.00                           | 3.90                            |

## Conclusion & Actionable Insights

### 1. Market Performance Segmentation

Product categories can be clearly segmented using customer ratings and engagement into four groups:

High Engagement – High Ratings: Strong demand with high customer satisfaction.

High Engagement – Low Ratings: Strong demand but weak customer experience.

Low Engagement – High Ratings: Good products with limited market exposure.

Low Engagement – Low Ratings: Weak demand and low customer trust.

This segmentation converts raw ratings into a decision-making framework rather than just descriptive metrics.

### 2. Growth & Investment Strategy

High Engagement – High Ratings (Scale)

Conclusion:

These categories show strong product–market fit and stable customer trust.

Actions:

Increase inventory allocation.

Prioritize these categories for predictable revenue growth.

Increase performance marketing spend.

#### High Engagement – Low Ratings (Fix First)

Conclusion:

Demand exists, but customer dissatisfaction points to execution or quality gaps.

Actions:

Analyze top negative reviews to identify key complaint drivers (durability, fit, compatibility).

Fix one core issue per category instead of broad changes.

Improve post-purchase experience before increasing traffic.

#### Low Engagement – High Ratings (Unlock Demand)

Conclusion:

These are underexposed but well-received products, indicating niche growth opportunities.

Actions:

Improve visibility through better search placement and category positioning.

Use educational content, influencers, or expert endorsements instead of discounts.

Focus on demand generation rather than product changes.

#### Low Engagement – Low Ratings (Deprioritize)

Conclusion:

These categories lack both demand and customer trust, posing a risk to brand perception.

Actions:

Stop paid promotions and performance marketing.

Investigate root causes through review classification (quality defects, missing parts, usability issues).

Avoid scaling until trust and product fundamentals improve.